

(1) For the x vs. y data to the right:

(a) use the least squares method for a linear equation, *viz.*, $y = ax + b$ using perpendicular offsets to determine the slope and intercept. Explain how you determined the correct root of the quadratic to use for the slope. Also, calculate the quality of the fit Q^2 .

(b) fit the same data to a straight line using vertical offsets and calculate the Q^2 for this fit.

(c) and finally, repeat part (b) for horizontal offsets.

Use the definitions from lecture to work this problem.

x	y
1	4.5
2	11
3	8
4	13.9
5	12
6	14
7	18
8	25
9	19
10	23.6

(2) (a) Find two different ways to linearize the equation $r = Az/(B + z)$ where A and B are adjustable constants. The reaction rate is r , and the concentration is z . Use both linearizations to fit the data to the right using perpendicular offsets. Compare the values of the constants for the two linearizations and the quality Q^2 of the fits.

(b) Use the solver on Excel to find a numerical solution for fitting the nonlinear equation to the data using vertical offsets. You will need a reasonable first guess for the solver in order to get the numerical method to converge. Use one of your results in part (a) for this guess. Compare this result with the two results in part (a).

z	r
2.50E+00	20
4.00E+00	43
1.00E+01	65
2.00E+01	106
4.00E+01	112
1.00E+02	128
2.00E+02	150
1.00E+03	165

(3) Murphy 1.2 Solve using matrices.

(4) Murphy 1.15

(5) Murphy 1.17